

Multiple Choice

1. **Answer:** D

Options: A) BaH₂; B) BaOH; C) Ba; D) BaO

Note: Correct option: D - BaO

2. **Answer:** A

Options: A) Blackbody radiation curves; B) Emission spectra of diatomic molecules; C) Electron diffraction patterns; D) Temperature dependence of reaction rates

Note: Correct option: A - Blackbody radiation curves

3. **Answer:** A

Options: A) B₁₂ icosahedra; B) B₈ cubes; C) B₆ octahedra; D) B₄ tetrahedra

Note: Correct option: A - B₁₂ icosahedra

4. **Answer:** D

Options: A) F; B) Si; C) O; D) Ca

Note: Correct option: D - Ca

5. **Answer:** D

Options: A) The orbitals are degenerate.; B) The set of orbitals has a tetrahedral geometry.; C) These orbitals are constructed from a linear combination of atomic orbitals.; D) Each hybrid orbital may hold four electrons.

Note: Correct option: D - Each hybrid orbital may hold four electrons.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '1'.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Unpaired electrons'.

Long Form Response

11. **Answer:** SCl₂. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** Arsenic-doped silicon. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key